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PREPARING ACTIVITY DLA-IS

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AIR FORCE-99 DLA

REVIEW AS,AT,AV,ER,MC,ME,SH YD 82

USER

PROJECT NUMBER 5365-0200

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MILITARY SPECIFICATION SHEET

TITLE

RING, RETAINING, EXTERNAL INTERLOCKING

SUPERSEDING MS90708B 25 AUG 89

AMSC-N/A FSC-5365

SPECIFICATION SHEET NUMBER

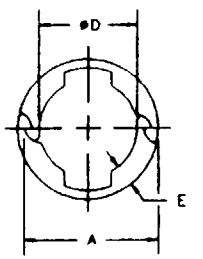
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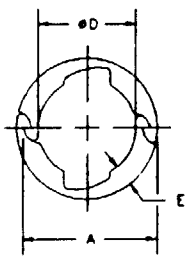
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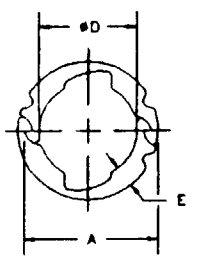
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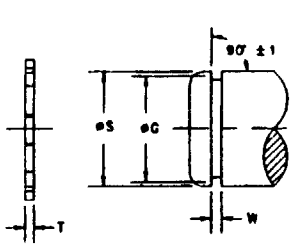



ALTERNATE DESIGN NO. 1 WITH TAPERED BRIDGE




ALTERNATE DESIGN NO. 2 WITH PLIER TIP NOTCHES TO FACILITATE ASSEMBLY



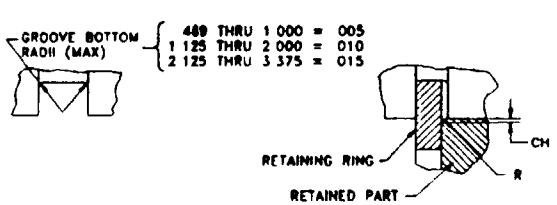




TABLE I DIMENSIONS ENLARGED DETAILS

| øS SHAFT (REF) | | øD[1] FREE (REF) | A WIDTH ACROSS LUGS | | E | | T[2] THICKNESS | | øG RECOMMENDED GROOVE (REF) | | W WIDTH | | R[4] / CH[4] OF RETAINED PART (REF) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| INCH | MM | BASIC | BASIC | TOL | BASIC | TOL | BASIC | TOL | BASIC | TOL | BASIC | TOL | MAX | MAX |
| 469 | 11 9 | 414 | 585 | ± 003 | 105 | | 035 | | 419 | ± 0015 | 040 | | 052 | 040 |
| 500 | 12 7 | 459 | 625 | | 105 | ± 005 | 035 | | 464 | 004 | 040 | | 052 | 040 |
| 594 | 15 1 | 538 | 710 | ± 004 | 105 | | 035 | | 544 | FIM[3] | 040 | + 003 | 052 | 040 |
| 625 | 15 9 | 569 | 740 | | 105 | | 035 | | 575 | | 040 | - 000 | 052 | 040 |
| 669 | 17 0 | 593 | 815 | | 135 | | 042 | | 599 | | 047 | | 065 | 050 |
| | | | | ± 005 | | | | | | ± 002 | | | | |
| 750 | 19 0 | 673 | 894 | | 135 | | 042 | ± 002 | 680 | 004 | 047 | | 065 | 050 |
| 781 | 19 8 | 703 | 925 | | 135 | | 042 | | 711 | FIM[3] | 047 | | 065 | 050 |
| 875 | 22 2 | 796 | 1 017 | | 135 | | 042 | | 805 | ± 003 | 047 | | 065 | 050 |
| 984 | 25 0 | 863 | 1 172 | | 188 | | 050 | | 872 | 004 | 056 | | 086 | 066 |
| 1 000 | 25 4 | 863 | 1 172 | ± 006 | 188 | ± 006 | 050 | | 872 | FIM[3] | 056 | | 081 | 062 |
| 1 125 | 28 6 | 1 002 | 1 311 | | 188 | | 050 | | 1 013 | | 056 | | 085 | 066 |
| 1 188 | 30 2 | 1 064 | 1 373 | | 188 | | 050 | | 1 075 | ± 003 | 056 | | 086 | 066 |
| 1 250 | 31 7 | 1 126 | 1 435 | | 188 | | 050 | | 1 138 | 005 | 056 | | 086 | 066 |
| 1 375 | 34 9 | 1 250 | 1 559 | ± 007 | .188 | | 050 | | 1 263 | FIM[3] | 056 | + 004 | 086 | 066 |
| 1 500 | 38 1 | 1 374 | 1 683 | | 188 | | 050 | | 1 388 | | 056 | - 000 | 086 | 066 |
| 1 562 | 39 7 | 1 412 | 1 778 | | 222 | | 062 | | 1 427 | | 068 | | 100 | 077 |
| 1 625 | 41 3 | 1 474 | 1 840 | | 222 | | 062 | | 1 489 | | 068 | | 100 | 077 |
| 1 750 | 44 4 | 1 597 | 1 963 | | 222 | | 062 | | 1 614 | ± 005 | 068 | | 100 | 077 |
| 1 772 | 45 0 | 1 597 | 1 963 | ± 008 | 222 | | 062 | | 1 614 | 005 | 068 | | 094 | 072 |
| 1 875 | 47 6 | 1 721 | 2 090 | | 222 | | 062 | | 1 739 | FIM[3] | 068 | | 100 | 077 |
| 1 969 | 50 0 | 1 779 | 2 209 | | 262 | | 078 | | 1 797 | | 086 | | 114 | 088 |
| 2 000 | 50 8 | 1 809 | 2 242 | | 262 | ± 007 | 078 | | 1 828 | | 086 | | 114 | 088 |
| 2 125 | 54 0 | 1 933 | 2 364 | | 262 | | 078 | | 1 953 | | 086 | | 114 | 088 |
| 2 156 | 54 8 | 1 933 | 2 364 | | 262 | | 078 | ± 003 | 1 953 | | 086 | | 104 | 080 |
| 2 250 | 57 1 | 2 057 | 2 488 | | 262 | | 078 | | 2 078 | | 086 | | 114 | 088 |
| | | | | ± 010 | | | | | | ± 005 | | | | |
| 2 375 | 60 3 | 2 180 | 2 610 | | 262 | | 078 | | 2 203 | 006 | 086 | + 005 | 114 | 088 |
| 2 500 | 63 5 | 2 304 | 2 734 | | 262 | | 078 | | 2 328 | FIM[3] | 086 | - 000 | 114 | 088 |
| 2 625 | 66 7 | 2 428 | 2 858 | | 262 | | 078 | | 2 453 | | 086 | | 114 | 088 |
| 2 750 | 69 8 | 2 518 | 3 049 | | 323 | | 093 | | 2 544 | | 103 | | 143 | 110 |
| 2 875 | 73 0 | 2 642 | 3 172 | | 323 | | 093 | | 2 669 | | 103 | | 143 | 110 |
| | | | | ± 012 | | | | | | ± 006 | | | | |
| 3 000 | 76 2 | 2 754 | 3 294 | | 329 | ± 008 | 093 | | 2 794 | 006 | 103 | | 143 | 110 |
| 3 250 | 82 5 | 3 013 | 3 542 | | 325 | | 093 | | 3 044 | FIM[3] | 103 | | 144 | 111 |
| 3 375 | 85 7 | 3 114 | 3 761 | ± 015 | 395 | | 109 | | 3 145 | | 120 | | 182 | 140 |

FOR FOOTNOTES SEE PAGE 2

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1/ ⌀D = DIMENSION "D" SHOULD NOT BE USED AS PART OF AN INSPECTION PROCEDURE USE DIMENSION A

2/ T = THICKNESS "T" APPLIES TO UNPLATED RINGS FOR CORROSION RESISTANT STEEL AND PLATED RINGS + 002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i e ± 002 SHOULD BE + 004/− 002

3/ FIM = (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT

4/ R AND CH = RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS THRUST LOADS FOR RINGS RETAINING PARTS WITH CORNER RADII OR CHAMFERS, ARE TABULATED ON PAGE 4

## REQUIREMENTS

1 CLASSIFICATION RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I CLASS B OF THE PROCUREMENT SPECIFICATION

2 MATERIAL

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568 OR ASTM A682

(b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700)

(c) BERYLLIUM COPPER ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194

3 HARDNESS

TABLE II HARDNESS

| ⌀SHAFT (REF) | CARBON STEEL | CORROSION RESISTANT STEEL | BERYLLIUM COPPER |
|---|---|---|---|
| 469 TO 625 INCL | 66 5-71 5HR30N | 44-51HRC | 56 5-62HR30N |
| 669 TO 1 500 INCL | 52 5-59 0HR45N | 44-51HRC | - |
| 1 562 TO 3 375 INCL | 48 0-53 0HRC | 44-51HRC | - |
| 669 TO 1 875 INCL | - | - | 37-43HRC |

4 PROTECTIVE FINISH OR SURFACE TREATMENT

(a) CARBON STEEL − SHALL BE AS SPECIFIED (SEE TABLE III OR IV)

© (1) CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5 OR ZINC-NICKEL PLATE IN ACCORDANCE WITH AMS 2417, TYPE 2

(2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn5, OR ASTM B695, TYPE II, CLASS 5

(3) PHOSPHATE COAT IN ACCORDANCE WITH DOD-P-16232, TYPE Z, CLASS 2

(b) CORROSION RESISTANT STEEL − SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH QQ-P-35

5 PART NUMBER THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV

EXAMPLE MS90708-1200 IS THE PART NUMBER FOR A CARBON STEEL CADMIUM PLATE EXTERNAL INTERLOCKING RETAINING RING FOR USE ON A 2 000 DIAMETER SHAFT

## NOTES

1 UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES

© 2 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE

© 3 UNLESS OTHERWISE SPECIFIED, ISSUES OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICITATION



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TABLE III DASH NUMBERS FOR MS90708

| ØS SHAFT (REF) | CARBON STEEL 1/ CADMIUM PLATE OR ZINC-NICKEL PLATE Ⓒ | CARBON STEEL 1/ ZINC COAT | CARBON STEEL 1/ PHOSPHATE COAT | STEEL CORROSION RESISTANT | BERYLLIUM 1/ COPPER |
|---|---|---|---|---|---|
| | DASH NO | DASH NO | DASH NO | DASH NO | DASH NO |
| 469 | -1046 | -2046 | -3046 | -4046 | -5046 |
| 500 | -1050 | -2050 | -3050 | -4050 | -5050 |
| 594 | -1059 | -2059 | -3059 | -4059 | -5059 |
| 625 | -1062 | -2062 | -3062 | -4062 | -5062 |
| 669 | -1066 | -2066 | -3066 | -4066 | -5066 |
| 750 | -1075 | 2075 | -3075 | -4075 | -5075 |
| 781 | -1078 | -2078 | -3078 | -4078 | -5078 |
| .875 | -1087 | -2087 | -3087 | -4087 | -5087 |
| 984 | -1098 | -2098 | -3098 | -4098 | -5098 |
| 1 000 | -1098 | -2098 | -3098 | -4098 | -5098 |
| 1 125 | -1112 | -2112 | -3112 | -4112 | -5112 |
| 1 188 | -1118 | -2118 | -3118 | -4118 | -5118 |
| 1 250 | -1125 | -2125 | -3125 | -4125 | -5125 |
| 1 375 | -1137 | -2137 | -3137 | -4137 | -5137 |
| 1 500 | -1150 | -2150 | -3150 | -4150 | -5150 |
| 1 562 | -1156 | -2156 | -3156 | -4156 | -5156 |
| 1 625 | -1162 | -2162 | -3162 | -4162 | -5162 |
| 1 750 | -1175 | -2175 | -3175 | -4175 | -5175 |
| 1 772 | -1175 | -2175 | -3175 | -4175 | -5175 |
| 1 875 | -1187 | -2187 | -3187 | -4187 | -5187 |
| 1 969 | -1196 | -2196 | -3196 | -4196 | |
| 2 000 | -1200 | -2200 | -3200 | -4200 | |
| 2 125 | -1212 | -2212 | -3212 | -4212 | |
| 2 156 | -1212 | -2212 | -3212 | -4212 | |
| 2 250 | -1225 | -2225 | -3225 | -4225 | |
| 2 375 | -1237 | -2237 | -3237 | -4237 | |
| 2 500 | -1250 | -2250 | -3250 | -4250 | |
| 2 625 | -1262 | -2262 | -3262 | -4262 | |
| 2 750 | -1275 | -2275 | -3275 | -4275 | |
| 2 875 | -1287 | -2287 | -3287 | -4287 | |
| 3 000 | -1300 | -2300 | -3300 | -4300 | |
| 3 250 | -1325 | -2325 | -3325 | -4325 | |
| 3 375 | -1337 | -2337 | -3337 | -4337 | |

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY OR IN FUEL OR LUBRICATION SYSTEMS OR WHEN USED AT TEMPERATURES OVER 450°F (233°C)

TABLE IV SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS)

| ØS SHAFT (REF) | INACTIVE | SUBSTITUTE | SUBSTITUTE | SUBSTITUTE |
|---|---|---|---|---|
| | CARBON STEEL | CARBON STEEL CADMIUM PLATE 1/ OR ZINC-NICKEL PLATE Ⓒ | CARBON STEEL ZINC COAT 1/ | CARBON STEEL PHOSPHATE COAT 1/ |
| | MS90708 | MS90708 | MS90708 | MS90708 |
| 469 | -46 | -1046 | -2046 | -3046 |
| 500 | -50 | -1050 | -2050 | -3050 |
| 594 | -59 | -1059 | -2059 | -3059 |
| 625 | -62 | -1062 | -2062 | -3062 |
| 669 | -66 | -1066 | -2066 | -3066 |
| 750 | -75 | -1075 | -2075 | -3075 |
| 781 | -78 | -1078 | -2078 | -3078 |
| 875 | -87 | -1087 | -2087 | -3087 |
| 984 | -98 | -1098 | -2098 | -3098 |
| 1 000 | -98 | -1098 | -2098 | -3098 |
| 1 125 | -112 | -1112 | -2112 | -3112 |
| 1 188 | -118 | -1118 | -2118 | -3118 |
| 1 250 | -125 | -1125 | -2125 | -3125 |
| 1 375 | -137 | -1137 | -2137 | -3137 |
| 1 500 | -150 | -1150 | -2150 | -3150 |
| 1 562 | -156 | -1156 | -2156 | -3156 |
| 1 625 | -162 | -1162 | -2162 | -3162 |
| 1 750 | -175 | -1175 | -2175 | -3175 |
| 1 772 | -175 | -1175 | -2175 | -3175 |
| 1.875 | -187 | -1187 | -2187 | -3187 |
| 1 969 | -196 | -1196 | -2196 | -3196 |
| 2 000 | -200 | -1200 | -2200 | -3200 |
| 2 125 | -212 | -1212 | -2212 | -3212 |
| 2 156 | -212 | -1212 | -2212 | -3212 |
| 2 250 | -225 | -1225 | -2225 | -3225 |
| 2 375 | -237 | -1237 | -2237 | -3237 |
| 2.500 | -250 | -1250 | -2250 | -3250 |
| 2 625 | -262 | -1262 | -2262 | -3262 |
| 2 750 | -275 | -1275 | -2275 | -3275 |
| 2.875 | -287 | -1287 | -2287 | -3287 |
| 3 000 | -300 | -1300 | -2300 | -3300 |
| 3 250 | -325 | -1325 | -2325 | -3325 |
| 3 375 | -337 | -1337 | -2337 | -3337 |

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY OR IN FUEL OR LUBRICATION SYSTEMS OR WHEN USED AT TEMPERATURES OVER 450°F (233°C)

| | | |
|---|---|---|
| PREPARING ACTIVITY DLA-IS | MILITARY SPECIFICATION SHEET | SPECIFICATION SHEET NUMBER |
| CUSTODIANS ARMY- AR NAVY- | TITLE | MS90708 28 JUNE 95 REV C |
| AIR FORCE- 99 DLA- | RING RETAINING EXTERNAL INTERLOCKING | SUPERSEDING |
| REVIEW AS AT,AV,ER,MC,ME,SH,YD,82 | | MS90708B 25 AUG 89 |
| USER | | AMSC- N/A FSC -5385 |
| PROJECT NUMBER 5365-0200 | | |
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RECOMMENDED DESIGN LIMITATIONS AND USAGE

(a) INTENDED USE - TO PROVIDE HIGH CIRCULAR SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS THE IDENTICAL SEMI-CIRCULAR HALVES HELD TOGETHER BY THE INTERLOCKING PRONGS FORM A BALANCED RING CONCENTRIC WITH THE SHAFT, WHICH WILL WITHSTAND HIGH ROTATIONAL SPEEDS THE USE OF THE FOLLOWING FORMULAS IS BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION

LIMITATION ON USE - THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC

WARNING - RING HALVES SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED (PROVIDING GROOVE HAS been MACHINED TO RECOMMENDED DIMENSIONS)

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V

TABLE V APPROXIMATE SAFETY RPM LIMITS

| ØSHAFT (INCHES) | | 500 | 1 000 | 1 500 | 2 000 | 2 500 | 3 000 | 3 375 |
|---|---|---|---|---|---|---|---|---|
| CARBON STEEL AND CORROSION RESISTANT STEEL | RPM LIMIT | 50,000 | 30 000 | 22 000 | 15,000 | 12,000 | 9,000 | 6,800 |
| BERYLLIUM COPPER | RPM LIMIT | 32,000 | 19,000 | 14,000 | | | | |

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS ABUTTING COMPONENTS TO HAVE SHARP CORNERS =

$$P = \frac{C_F \pi S T X}{F}$$

WHERE

P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY, F = 3 IS RECOMMENDED TO INSURE A SAFE WORKING LOAD
$C_F$ = CONVERSION FACTOR, $C_F$ = 3/4 IS RECOMMENDED SINCE SHEAR AREA IS REDUCED AT THE CUTOUTS IN THE RING

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL =

$$P = \frac{C_F \pi S d Y}{F}$$

WHERE

P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL
F = FACTOR OF SAFETY, F = 2 IS RECOMMENDED TO INSURE A SAFE WORKING LOAD
$C_F$ = CONVERSION FACTOR, $C_F$ = 3/4 IS RECOMMENDED SINCE CONTACT AREA IN THE GROOVE WALL IS REDUCED AT THE CUTOUTS IN THE RING

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$$Z = 3d$$

WHERE

Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

(e) ALLOWABLE SHAFT DIAMETER =

$$S = \sqrt{G^2 + \frac{4FP}{Y \pi C_F}}$$

WHERE

S = ALLOWABLE SHAFT DIAMETER
G = GROOVE DIAMETER
F = FACTOR OF SAFETY - SEE FORMULA (c) ABOVE
P = DESIGN LOAD
Y = YIELD STRENGTH IN COMPRESSION OF GROOVE MATERIAL (PSI)
$C_F$ = CONVERSION FACTOR, SEE FORMULA (c) ABOVE

1/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL
X = 110 000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER

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CUSTODIANS ARMY-AR NAVY-
AIR FORCE - 99 DLA-
REVIEW AS AT AV ER,MC ME SH YD 82
USER
PROJECT NUMBER 5365-0200

MILITARY SPECIFICATION SHEET
TITLE
RING RETAINING EXTERNAL INTERLOCKING

SPECIFICATION SHEET NUMBER
MS90708 28 JUNE 95 REV C
SUPERSEDING
MS907088 25 AUG 89
AMSC - N/A FSC 5365

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(f) DIFFERENTIAL ROTATION =

DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RING UP TO THE ALLOWABLE LOAD CALCULATED TO THE ABOVE FORMULA (b)

(g) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ = FOR THE RING (INCH POUNDS) ABUTTING COMPONENTS TO HAVE SHARP CORNERS

$I_G = \frac{Pd}{2}$ = FOR THE GROOVE (INCH POUNDS)

WHERE
- P = ALLOWABLE THRUST LOAD OF RING OR GROOVES (POUNDS)
- T = RING THICKNESS (INCHES)
- $I_G$ = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
- d = GROOVE DEPTH (INCHES)
- $I_R$ = IMPACT CAPACITY OF RING (INCH POUNDS)

(h) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED, WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON PAGE 4 WILL APPLY THE CORNER RADII AND CHAMFERS LISTED ON PAGE 1 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELAED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE VI IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY IN ACCORDANCE WITH THE FOLLOWING FORMULAS

$P^1 = \frac{P\ CH}{CH^1}$ or $P^1 = \frac{P\ R}{R^1}$

WHERE
- $P^1$ = NEW ALLOWABLE THRUST LOAD
- P = LISTED ALLOWABLE THRUST LOAD
- $CH^1$ = NEW (SMALLER) CHAMFER
- CH = LISTED CHAMFER
- $R^1$ = NEW (SMALLER) CORNER RADIUS
- R = LISTED CORNER RADIUS

LIMIT LOADS LISTED BELOW ARE BASED ON RINGS OF STEEL (WORKING STRESS 250,000 PSI) AND OF BERYLLIUM COPPER (WORKING STRESS 180,000 PSI) IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA GIVEN ABOVE ARE LESS, THEN THEY SHOULD BE USED

TABLE VI LIMIT LOADS

| NOMINAL RING SIZE | | ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS | |
|---|---|---|---|
| FROM | TO | CARBON STEEL OR CORROSION RESISTANT STEEL | BERYLLIUM COPPER |
| 469 | 625 | 610 LB | 440 LB |
| 669 | 875 | 880 LB | 630 LB |
| 984 | 1.500 | 1250 LB | 900 LB |
| 1 562 | 1 875 | 1900 LB | 1370 LB |
| 1 969 | 2 625 | 3050 LB | – |
| 2 750 | 3 250 | 4300 LB | – |
| 3 375 | – | 5950 LB | – |

PREPARING ACTIVITY DLA-IS
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PROJECT NUMBER 5365-0200

MILITARY SPECIFICATION SHEET
TITLE
RING RETAINING, EXTERNAL INTERLOCKING

SPECIFICATION SHEET NUMBER
MS90708 28 JUNE 95 REV C
SUPERSEDING
MS907088 25 AUG 89
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